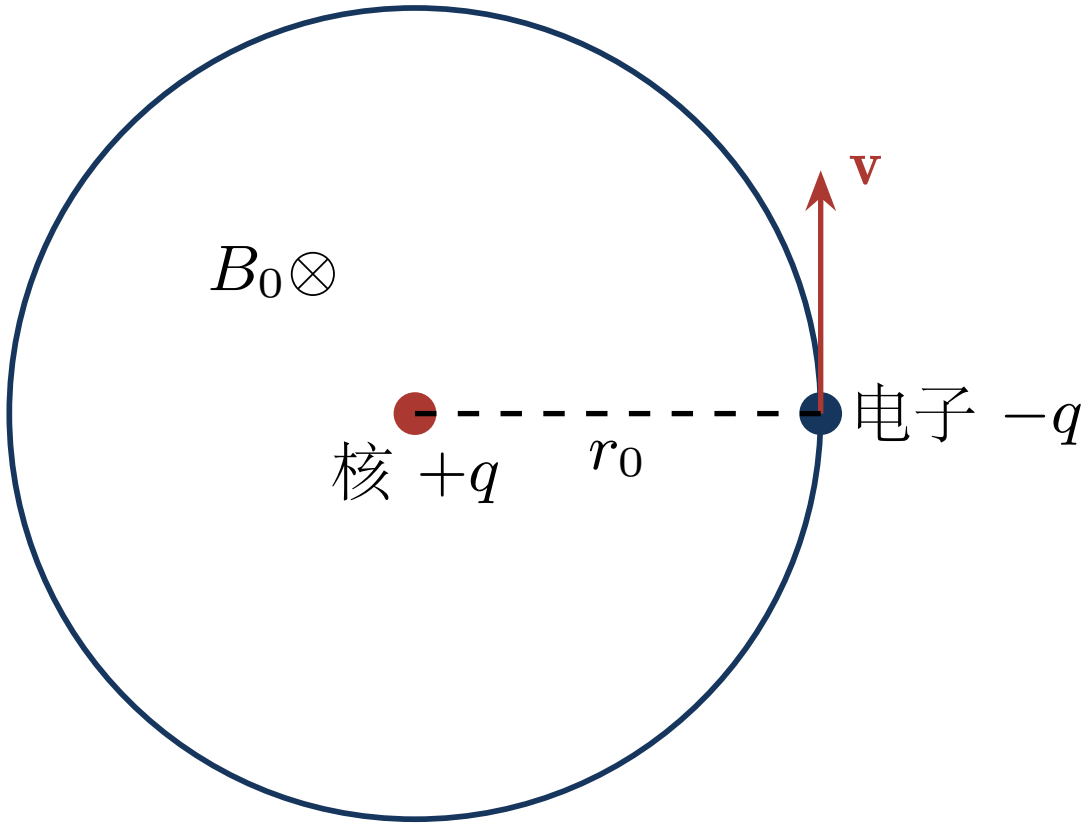
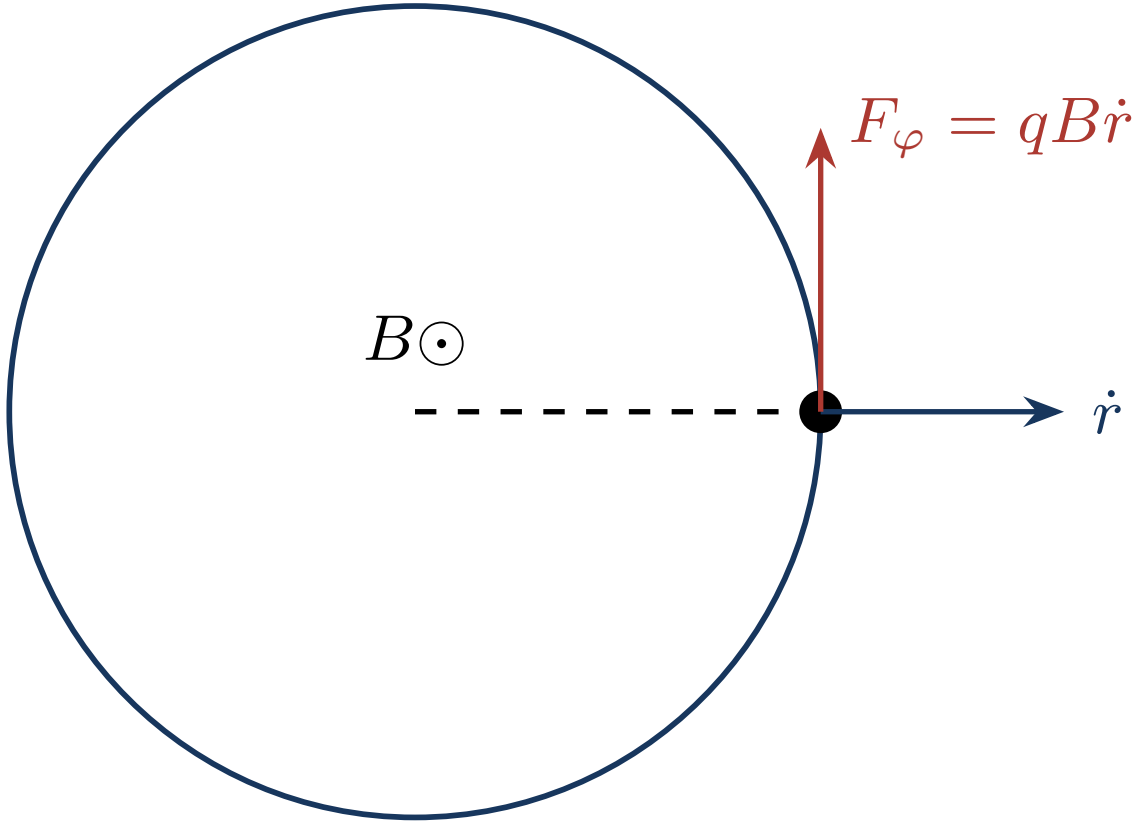
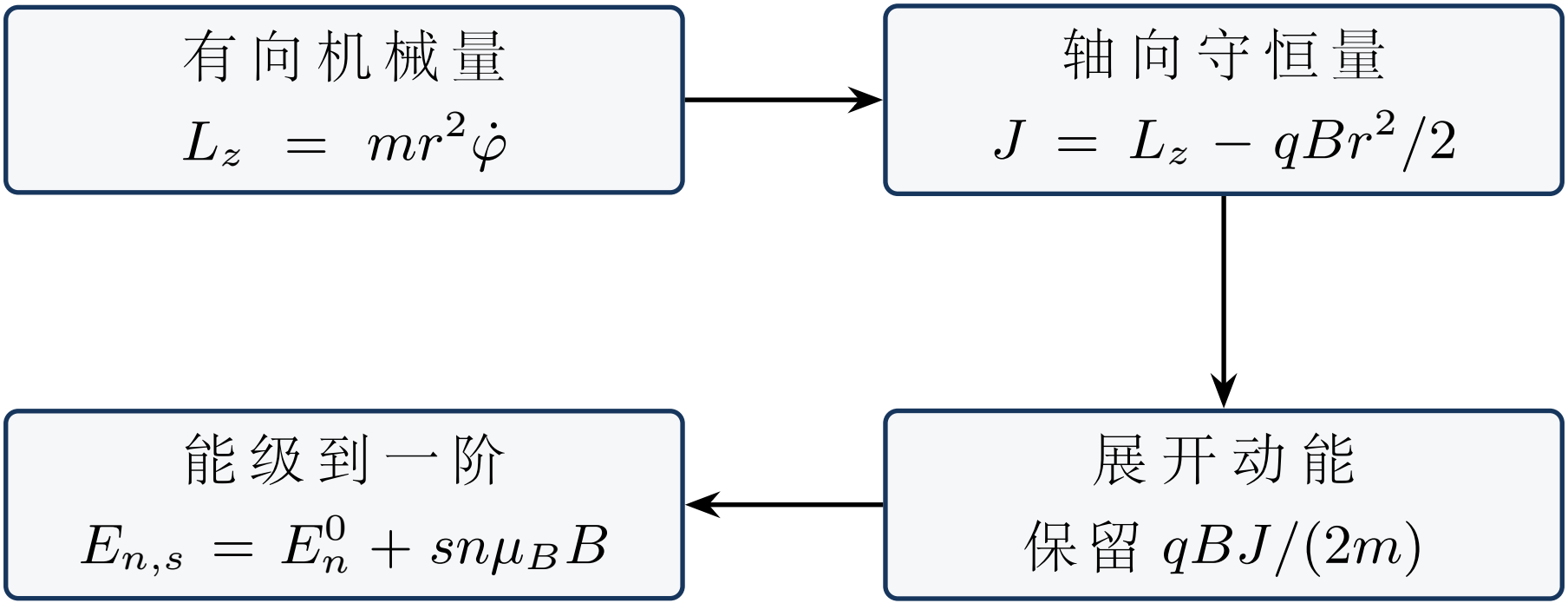


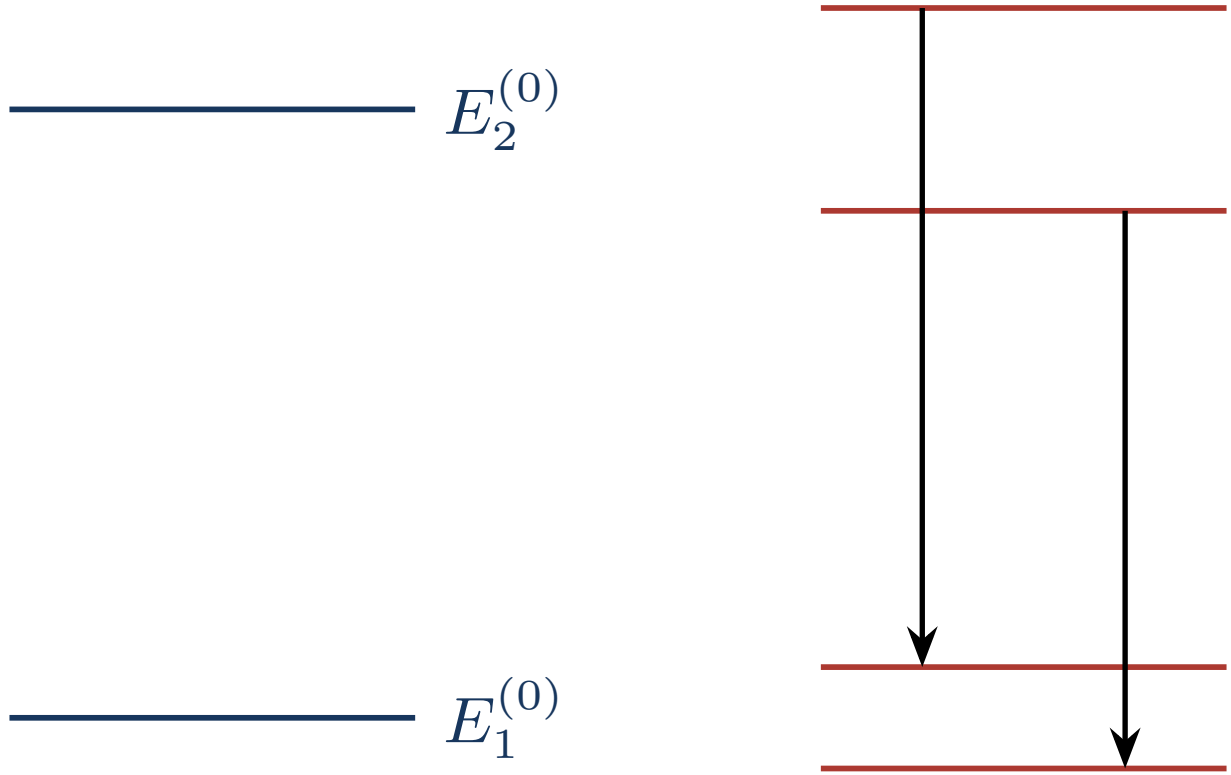
电子逆时针，等效电流顺时针，核处自生磁场向纸内



外场向纸外，径向速度产生切向磁力；$\dot{L}_z = qBr\dot{r}$



求半径时常量项导数为零，求能量时不能删除



无场　　弱场示意

教学圆轨道模型的同绕行方向分支

$\Delta\nu = \pm\mu_B B/h$；分裂尺度夸大以示意

半径方程

$J^2 = mkr + q^2B^2r^4/4$

首修正为 B^2

能量展开

$qBJ/(2m)$

已有 B 的首修正

半径无一阶变化，不等于光谱无一阶变化；保持近似精度一致